



ROOT CAUSE UNLOCKED · CONDITION KIT

The Detox Root Cause *Kit*

Which tests are real, which one to refuse, and exactly what to ask your doctor for

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The Detox Root Cause Kit

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BEFORE YOU START

Contents

Read this part first

The eight keys, and what tests each one

Lead: blood, and know the reference value

Mercury: match the specimen to the exposure

Arsenic: the seafood trap that produces most “high” results

Cadmium: two tests, two time windows

Functional ranges for the markers in this guide

Reading your results as a pattern

The conversation with your doctor

What these labs cannot tell you

When to seek clinical review

Your next step

References

Read this part first

The one idea underneath everything

Why this job falls to you in the first place

Fish: the highest-yield single swap

Packaging and reheating: the plastic you can actually skip

Water: know your source, filter accordingly

The organic question, answered honestly

Crucifers and broccoli sprouts: the best-proven detox food on earth

Protein: the raw material Phase II runs on

Color, alliums, and the antioxidant supply line

B vitamins and the methylation route

Fiber: the one nutrient with direct evidence against a “forever chemical”

Hydration: the kidney’s half of the deal

Movement and sweat, graded honestly

What a day looks like on this pattern

Working with Root Health

References

Read this part first

Why nutrient evidence looks thinner than it is

Start with the uncomfortable fact about this product category

Glutathione: the honest version of the strongest story

NAC: the precursor with a pharmacy pedigree

Sulforaphane: prefer the food, and here is the honest supplement note

Milk thistle (silymarin)

Chlorella, cilantro, and the “gentle binder” shelf

Activated charcoal

Chelation without a confirmed diagnosis

High-dose selenium

High-dose vitamin C, for one specific group

What I deliberately left out

Where to find quality versions of these

Working with Root Health

References

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Some situations skip this guide entirely. Suspected acute poisoning, a known ingestion, a sudden occupational exposure, a child who may have eaten paint chips or a metal object, or any exposure concern during pregnancy or breastfeeding: call Poison Control at 1-800-222-1222 or seek immediate medical care. Testing strategy conversations are for a calm day. Those situations are not a calm day.

Children and pregnancy get a physician from the start. Children absorb lead far more efficiently than adults, and pregnancy mobilizes lead stored in bone years earlier. Nothing in this guide replaces pediatric or obstetric care.

Nothing here is a reason to stop a prescription on your own.

The eight keys, and what tests each one

Unlock Your Detox Capacity is organized around eight keys. Here is how each maps to something measurable, and, just as importantly, where no useful test exists.

Key	What it means	How you find it
1. Toxic burden is a spectrum	A hypothesis to test, not a diagnosis to assume	The exposure audit, before any lab
2. Where exposure comes from	Water, food, home, work, hobbies	The audit again. No blood test replaces it
3. Elimination pathways	Liver, bile, kidneys, gut, lymph, sweat	Liver enzymes, kidney markers, bowel habits
4. The nutrients pathways need	Raw material for Phase I and II	RBC glutathione, zinc, selenium, B-vitamin status
5. Testing honestly	Unprovoked yes, provoked no	This entire guide
6. Reducing exposure	The highest-yield lever	Not a lab. The audit and your habits
7. The oxalate question	Stones are real, the syndrome is a hypothesis	24-hour urine oxalate, for stone formers
8. Conservative drainage	Open the drain before the faucet	Bowel regularity, hydration. Not a blood test

Two of the most important items on that list are not blood tests. The exposure audit tells you more than most panels, and it costs nothing.

The one decision that matters more than any single result: unprovoked versus provoked

If you remember one thing from this guide, make it this one.

Unprovoked testing measures blood or urine collected under normal conditions and compares it against reference ranges built the same way. This is the validated, defensible approach used in clinical toxicology and

population research.

Provoked or “challenge” testing gives a chelating drug first, then collects urine and compares it against those same unprovoked reference ranges. That comparison is not legitimate: a chelator pulls stored metal into circulation and elevates the result in almost anyone, there is no validated reference range for post-chelator urine, and the professional toxicology literature recommends against the practice (PMID 24113861).

How badly does provoked testing perform when actually checked? In a prospective cohort of patients referred for medical toxicology evaluation, a positive provoked urine result had a positive predictive value of about 4 percent for actual heavy metal toxicity confirmed by a medical toxicologist (PMID 34184587). In plain terms, roughly 96 out of 100 positive results pointed at a toxicity the patient did not have.

And the provoking agent is not harmless. Chelation drugs are real pharmaceuticals. Wrong-agent errors have caused documented deaths from sudden hypocalcemia (PMID 16882789). A test that risks the drug’s harms to produce a number that means almost nothing is a test worth declining politely and firmly.

What to say if it is offered: “I would prefer unprovoked blood and urine testing interpreted against standard reference ranges.” That sentence is accurate, reasonable, and hard to argue with.

Heavy metals, metal by metal

The general framing for all four: these are established toxicants with well-described sources and validated unprovoked tests (PMID 14757716). A detectable level is not automatically a dangerous level. The question is always the level, the trend, and the source.

Lead: blood, and know the reference value

The test is a blood lead level. Lead in blood reflects recent and ongoing exposure plus some release from bone stores.

The number to know: 3.5 micrograms per deciliter. That is the CDC blood lead reference value, updated in 2021, set at the 97.5th percentile of U.S. children aged 1 to 5. It is a prompt for finding and removing the source, not a treatment threshold, and the CDC states plainly that there is no known safe blood lead level (PMID 34710078).

If a child in your home has any plausible exposure (a pre-1978 home, renovation dust, imported ceramics or spices, certain hobbies), blood lead testing runs through your pediatrician, not through a wellness panel.

Mercury: match the specimen to the exposure

Blood mercury mainly reflects methylmercury from fish, the dominant route for most people. It reflects the last month or two of intake, so a high-fish week before the draw shows up in the number.

Urine mercury mainly reflects inorganic and elemental mercury, the occupational and dental-amalgam forms. Testing urine to assess a fish-heavy diet, or blood to assess workplace vapor exposure, measures the wrong compartment.

Arsenic: the seafood trap that produces most “high” results

Urine is the standard specimen for arsenic, and it carries a trap worth knowing before you draw it. Seafood contains organic arsenic compounds, arsenobetaine chief among them, that are essentially nontoxic but massively inflate a total urine arsenic result. In a national U.S. sample, people who had eaten seafood in the prior 24 hours had median total urine arsenic more than three times higher than those who had not (PMID 21093857).

The two fixes, either works: avoid seafood for several days before an unprovoked urine arsenic test, or ask for speciated arsenic, which separates the harmless seafood forms from the inorganic forms that matter. A “high arsenic” result on a total, unspeciated test after a sushi dinner is the single most common false alarm in this territory.

Cadmium: two tests, two time windows

Blood cadmium reflects recent exposure; urine cadmium reflects cumulative body burden, because the kidney is where cadmium accumulates over decades. Smoking is the dominant source in smokers, food in everyone else. If you have never smoked and have no occupational exposure, an elevated result deserves a repeat before it deserves a reaction.

PFAS: the blood test that recently became askable

PFAS (“forever chemicals”) persist in the body for years, which is exactly why a blood test is informative: it integrates years of exposure rather than yesterday’s (PMID 30470793).

Testing is a serum PFAS panel, now available through major laboratories. The National Academies of Sciences, Engineering, and Medicine issued clinical guidance in 2022 recommending that clinicians offer testing to patients with likely elevated exposure (contaminated water systems, certain occupations, communities near known sources), and it organized follow-up into tiers by the summed blood level: below 2 nanograms per milliliter, no adverse effects expected; 2 to 20, standard clinical follow-up with attention to the conditions PFAS are associated with; above 20, additional screening conversation with your physician. That is institutional guidance rather than a single study, and it is the frame most laboratories now report against.

What the result changes: mostly exposure hunting. There is no approved medication that removes PFAS, so a high result points back at water, packaging, and occupational sources, the Key 6 territory where the real leverage lives.

The panel you already have: what your liver and kidney markers say about elimination

Before any specialty testing, the standard chemistry panel already reads on the two organs doing most of the elimination work.

ALT and AST are the basic liver injury markers. ALT is the more liver-specific of the two; an elevated AST with a normal ALT is more often muscle than liver.

GGT deserves more attention than it gets in this specific context. It is the most sensitive routine hepatic enzyme marker for oxidative stress, and its half-life of two to three weeks means it reflects the recent past rather than this morning. Population research has associated background-level persistent organic pollutant

burden with shifts in liver markers, within ranges most reports would call normal (PMID 25173059). A GGT drifting upward across the functional range, with normal ALT and AST, is a reasonable prompt to look at oxidative and biliary load, including alcohol honestly counted.

Bilirubin is worth a glance because bile is a primary exit route: a persistently elevated result needs its direct and indirect fractions separated by your physician, since the two point at different causes.

Creatinine, BUN, and eGFR read the kidney, the other main exit. eGFR above 90 is normal kidney function, and persistent values below 60 define chronic kidney disease, which is physician territory and changes what is safe to take, chelation and high-dose anything included. Creatinine runs naturally higher in muscular people, so trend beats single value here too.

Uric acid is included in the table below mostly as a metabolic context marker: fructose intake, not protein, is its main dietary driver, and it travels with the metabolic picture that also burdens the liver.

Capacity markers: measuring the raw material

RBC glutathione reflects intracellular glutathione status, which plasma glutathione is too transient to capture. Glutathione is the master Phase II conjugate, assembled from glutamate, cysteine, and glycine, and depleted fastest under oxidative load. A low result is a resource problem you can actually address with food and, where appropriate, supplementation, which is what the companion supplement guide covers.

Zinc and selenium are cofactors across the antioxidant and conjugation systems. Two honest caveats from our own interpretive framework: plasma zinc drops during any acute illness because zinc redistributes to the liver, so do not draw it sick; and serum selenium reflects recent intake more than tissue status. Selenium also has a genuinely narrow window, and supplementing above documented need is not a favor to yourself.

Homocysteine, if it is on your panel, is a methylation cycle marker, and methylation is one of the Phase II conjugation routes. Read an elevated result as a signal to investigate B12, folate, B6, thyroid, and kidney status, not as a toxicity meter and not as a cardiac target proven to change outcomes on its own.

The oxalate tests, honestly

If you form calcium oxalate kidney stones, a 24-hour urine collection including oxalate is standard, validated, and genuinely useful: urinary oxalate tracks stone risk in a graded way, and roughly half of it comes from diet in normal individuals (PMID 33379176). That result is worth acting on with the diet guide's oxalate section.

If you do not form stones, there is no validated lab that diagnoses a broader "oxalate toxicity syndrome," because that syndrome is a community hypothesis rather than an established diagnosis. Organic acid panels sold in wellness settings report oxalate metabolites, but no validated interpretive framework connects those

values to the symptom picture the hypothesis describes. You are allowed to find the hypothesis interesting. Just do not pay for a number that cannot answer the question.

Functional ranges for the markers in this guide

Clinical targets I use at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose. Heavy metal and PFAS results are interpreted against their own laboratory and CDC reference frameworks described above, not against this table.

Marker	Units	Functional target	Approximate standard range
ALT	IU/L	10 to 26	up to 40
AST	IU/L	10 to 26	up to 40
GGT	IU/L	9 to 17	up to 55
Total bilirubin	mg/dL	0.4 to 0.9	up to 1.2
Creatinine	mg/dL	0.8 to 1.1	0.5 to 1.3
BUN	mg/dL	10 to 16	5 to 25
Uric acid	mg/dL	3.5 to 5.5	2.5 to 6.5
Glutathione (RBC)	umol/gHb	7 to 12	5 to 15
Zinc (plasma)	ug/dL	70 to 110	60 to 120
Selenium (serum)	ug/L	85 to 140	60 to 160
Homocysteine	umol/L	under 7.0	under 10 to 15, varies by lab

Reading your results as a pattern

Pattern one: everything normal, symptoms real. The most common outcome of honest testing. It does not mean nothing is wrong; it means “toxins” is not the supported explanation, and the honest next step is looking where the evidence points instead. A guide that cannot say this is not being straight with you.

Pattern two: liver markers drifting, no specific toxicant found. Look at alcohol honestly counted, medications and supplements (the supplement guide’s liver-injury section applies), sleep, and the metabolic picture, before exotic explanations.

Pattern three: a genuinely elevated metal on unprovoked testing. This is a finding, and it goes to your physician, with source-hunting from the exposure audit as the parallel track. Confirmed, clinically significant toxicity is medical toxicology territory, where chelation is a real treatment with real monitoring, not a wellness protocol.

Pattern four: low capacity markers. Low RBC glutathione, low zinc, marginal selenium. This is the most actionable pattern in the guide and the one the diet and supplement guides are built for.

The conversation with your doctor

A request list you can bring

- Blood lead level, if exposure history suggests it
- Blood mercury (fish exposure) or urine mercury (occupational), matched to your actual exposure
- Speciated urine arsenic, or several seafood-free days before a total arsenic test
- Blood and urine cadmium if there is a smoking or occupational history
- Serum PFAS panel if your water system, occupation, or community suggests elevated exposure
- The standard panel read with elimination in mind: ALT, AST, GGT, bilirubin, creatinine, BUN, eGFR
- RBC glutathione, plasma zinc, serum selenium
- 24-hour urine oxalate, if you have ever formed a kidney stone
- **Decline provoked or challenge testing**, in exactly those words

The sentence that helps

“I would like unprovoked testing matched to my actual exposure history, interpreted against standard reference ranges.” It signals that you know the territory, and it steers the visit toward the testing that means something.

What these labs cannot tell you

- **Whether a symptom is caused by a toxin.** Association and causation separate here exactly as they do everywhere else in this product line.
- **Your lifetime burden.** Most tests read a recent window; only some (urine cadmium, serum PFAS) integrate years.
- **Whether you need chelation.** That is a medical toxicology decision built on confirmed, clinically significant results, never on a wellness panel.

When to seek clinical review

Emergency, repeated from the top: suspected acute poisoning, a known ingestion, a child with plausible lead exposure, or exposure concerns in pregnancy. Poison Control at 1-800-222-1222, or immediate medical care.

Soon: any confirmed elevated metal on unprovoked testing, an ALT or GGT that stays elevated on repeat, an eGFR below 60, or new neurological symptoms with a plausible exposure history.

Your next step

The companion guides do the rest: **The Detox Diet Guide** covers reducing what comes in and feeding the pathways that move it out, and **The Detox Supplement Guide** covers what genuinely helps, what is merely popular, and which detox products injure the organ they claim to support.

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[utm_source=guide&utm_medium=ebook&utm_campaign=detox-labs](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=detox-labs)***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

***Get in touch: [rootcauseunlocked.com/contact?](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=detox-labs)
[utm_source=guide&utm_medium=ebook&utm_campaign=detox-labs](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=detox-labs)***

If you would rather start on your own, start with the exposure audit in *Unlock Your Detox Capacity* and two weeks of honest tracking:

***Free tracker: [rootcauseunlocked.com/tracker?](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=detox-labs)
[utm_source=guide&utm_medium=ebook&utm_campaign=detox-labs](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=detox-labs)***

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All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in *Detox-Kit-Citation-Ledger.md*. The National Academies 2022 PFAS clinical guidance is an institutional report rather than a journal article and is cited by name above.

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Educational content only. Not medical advice. Not a diagnosis. Not a reason to stop or refuse any medication. Suspected acute poisoning is an emergency: Poison Control 1-800-222-1222.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Suspected acute poisoning, a known ingestion, a child with plausible lead exposure, or exposure concerns in pregnancy or breastfeeding are emergencies, not diet projects. Poison Control: 1-800-222-1222.

This guide contains no cleanse, no fast, and no flush. If you bought it hoping for one, read the section near the end on why, because that reasoning is worth more than the cleanse would have been.

Nothing here is a reason to stop a prescription on your own. If you have kidney disease or a history of kidney stones, the oxalate and hydration sections carry specific cautions for you.

The one idea underneath everything

Key 6 of *Unlock Your Detox Capacity* makes the argument this whole guide rests on: **you cannot out-supplement continuous reloading**. Food and water are simultaneously the biggest source of ongoing exposure and the biggest lever for supporting elimination. That makes the plate the single most important piece of detox territory you control, from both directions at once.

So this guide works both directions: part one reduces what is coming in, part two feeds the pathways that move things out, part three keeps the exits open.

Part one: reduce what is coming in

Why this job falls to you in the first place

The European Union generally requires a food ingredient to be shown safe before it reaches the market. The United States runs the opposite default: substances stay on the market until harm is proven, and under the “generally recognized as safe” pathway a company can even self-certify a new ingredient without notifying the FDA. The results are visible on the label. Potassium bromate remains legal in American bread decades after the EU, the UK, and Canada banned it. Titanium dioxide left EU food in 2022 and stayed in ours. Red Dye No. 3 was barred from US cosmetics in 1990 over cancer findings and remained in food until the FDA finally revoked its authorization in 2025. (These are public FDA and European Commission regulatory records, cited by name.)

Notice the double standard this creates, because it runs through everything in this territory: a single trial showing harm from a vitamin generates immediate warnings, while decades of evidence against a profitable additive generates “more research is needed.” None of that is a conspiracy. It is a default setting, plus money. But it means the American system will not do your exposure reduction for you, and waiting for a ban is not a plan. That is why this section exists, and why the audit beats outrage: you can remove most of these exposures from your own kitchen years before any agency does it for you.

Fish: the highest-yield single swap

Fish is the main mercury route for most people, and the fix is not eating less fish. It is choosing smaller, shorter-lived species. Salmon, sardines, trout, and herring carry meaningfully less mercury than swordfish, shark, king mackerel, and large tuna, while carrying more omega-3 per serving in most cases. Same habit, better inputs.

If you are pregnant, planning pregnancy, or feeding young children, follow the joint FDA and EPA fish advice categories, which exist for exactly this reason. That is standing public health guidance, and this guide defers to it.

Packaging and reheating: the plastic you can actually skip

The measurable plastic and PFAS exposure most people can reduce is the food-contact kind: heavily packaged foods, grease-resistant wrappers, and heating food in plastic (PMID 30470793). Three habits do most of the work: transfer food out of plastic before reheating, favor glass or stainless containers for hot and fatty foods (both heat and fat increase migration), and let the share of your food that arrives in a wrapper drift down. None of this requires purity. Direction beats perfection here.

Water: know your source, filter accordingly

If your water comes from a private well, it has never been tested unless you tested it: arsenic and lead are the classic well findings, and testing is cheap relative to guessing. If you are near a known PFAS contamination site, reverse osmosis is the most consistently effective home treatment for PFAS specifically, with combined carbon filtration as a reasonable second choice (PMID 30470793). If neither applies, your annual municipal water quality report tells you what is actually in the water, which beats buying a filter for a problem you may not have.

The organic question, answered honestly

Choosing organic measurably lowers urinary pesticide metabolite levels; that part is documented. Whether that reduction changes health outcomes has not been demonstrated, and this guide will not claim it has. A reasonable, budget-aware approach: if the cost matters, spend the organic premium on the produce categories with the highest residue burden and buy the rest conventional. Washing and peeling reduce residue on either. What is not reasonable is letting pesticide worry reduce how many vegetables you eat, because the vegetable itself is doing provable work in part two and the residue is doing hypothetical harm.

Part two: feed the pathways

Crucifers and broccoli sprouts: the best-proven detox food on earth

This is the rare wellness claim with randomized human trials behind it, and they are worth describing properly.

Broccoli sprouts are an exceptionally concentrated source of glucoraphanin, which the body converts to sulforaphane, a potent inducer of the Phase II enzyme system that conjugates and clears reactive compounds (PMID 9294217). In Qidong, China, a region with heavy air pollution, a randomized placebo-controlled trial of a broccoli sprout beverage found significantly increased urinary excretion of the glutathione-derived conjugates of benzene (61 percent higher) and acrolein (23 percent higher), beginning rapidly and sustained across the 12-week trial (PMID 24913818). A follow-up randomized dose-ranging trial found the effect held at reduced doses, with benzene conjugate excretion still significantly elevated at one-fifth of the original dose (PMID 31268126).

Read what that is: real human beings, randomized, measurably excreting more of a real airborne carcinogen because of a food. Not a mouse study, not a marketing claim.

How to use it: cruciferous vegetables most days (broccoli, cauliflower, Brussels sprouts, kale, arugula, cabbage), with broccoli sprouts as the concentrated option. Sprouts are cheap to grow on a windowsill in about five days, need no cooking, and go into salads, smoothies, and sandwiches. Chewing matters: the converting enzyme is released by chopping or chewing, so a swallowed-whole sprout does less work.

Protein: the raw material Phase II runs on

Every conjugation route in Phase II consumes amino acids: glycine, cysteine, glutamine, taurine, methionine. Glutathione itself is built from three of them. The practical translation is unglamorous: adequate protein at every meal, roughly a palm-sized portion, from eggs, fish, poultry, meat, dairy, legumes, or a combination. Collagen-rich and whole-animal cuts add glycine specifically. A “detox” week of juice and salad removes precisely the raw material the system needs most, which is part of why cleanses backfire.

Color, alliums, and the antioxidant supply line

Phase I generates reactive byproducts as a normal cost of doing business, and the antioxidant network absorbs them: vitamin C and E, selenium, zinc, and the polyphenol variety that comes with genuinely colorful plates. Garlic and onions add sulfur compounds that feed the same sulfation and glutathione chemistry the crucifers support. The instruction is the least original one in nutrition and still correct: many colors, most days, mostly plants, plus the allium family regularly.

B vitamins and the methylation route

One of Phase II's conjugation routes is methylation, which runs on B12, folate, and B6. Eggs, leafy greens, legumes, meat, and fish cover the territory; a basic B-complex is a reasonable insurance policy when intake is inconsistent, and the supplement guide covers when testing justifies more than that.

Part three: keep the exits open

Fiber: the one nutrient with direct evidence against a “forever chemical”

Compounds your liver exports into bile can be reabsorbed from the gut if they sit there, a loop called enterohepatic recirculation. Fiber interrupts the loop and carries the traffic out. This is not just mechanism talk: in national U.S. survey data, higher fiber intake was associated with measurably lower serum concentrations of several PFAS, most plausibly through exactly this route (PMID 33395939).

Target: 25 to 35 grams daily, built gradually. Beans and lentils, oats, chia and flax, vegetables, fruit with skins, whole grains. Gradually matters: jumping from 12 grams to 35 overnight produces a week of misery and a quit. Add roughly five grams every few days and drink water alongside.

Bowel regularity is the metric that matters: once or twice daily, comfortably. If that is not your baseline, fiber, water, movement, and magnesium (covered in the supplement guide) are the levers, and fixing this comes before any thought of mobilizing anything.

Hydration: the kidney's half of the deal

The kidneys clear water-soluble compounds continuously, and they do it better with adequate volume. Pale-straw urine is the practical gauge. If you form kidney stones, hydration is doubly non-negotiable and your target fluid intake likely came from your urologist; honor that number.

Movement and sweat, graded honestly

Lymph has no pump; it moves when you move. Daily walking, and any exercise that makes you breathe deeply, is doing quiet drainage work that no product replaces. Sweating is a real but smaller elimination route: a small human monitoring study found certain toxic elements detectable in sweat, in some cases when blood levels were undetectable (PMID 21057782). That is preliminary data from a small study, not proof that sauna clears meaningful burden, and PFAS in particular do not leave through sweat. Enjoy sauna if you enjoy sauna; hydrate around it; do not assign it a job it has not been shown to do.

The oxalate section: for the readers it actually applies to

If you form calcium oxalate kidney stones, this is for you. Dietary oxalate contributes roughly half of urinary oxalate, and urinary oxalate tracks stone risk in a graded way (PMID 33379176). The evidence-supported moves: pair calcium with meals (calcium binds oxalate in the gut so it exits in stool rather than being absorbed; normal dietary calcium, not restriction, is the stone-preventive position), hydrate seriously, moderate the heaviest oxalate foods (spinach, rhubarb, almonds, beets, chocolate), and be aware that high-dose vitamin C supplementation is a documented precursor to the body's own oxalate production (PMID 30566003).

If you do not form stones and have no confirmed elevated urinary oxalate, a strict low-oxalate diet has no established indication, and the broader “oxalate toxicity” symptom syndrome remains a community hypothesis, as the ebook’s Key 7 lays out. If you choose to reduce oxalate anyway, change gradually over several weeks rather than abruptly, and do not let a hypothesis crowd out vegetables with documented benefits.

What this guide deliberately does not include, and why

No juice cleanses or multi-day fasts. Phase I and II run on protein, B vitamins, and antioxidants. Prolonged severe restriction removes the raw material at the exact moment marketing claims the system is working hardest. A cleanse feels like something is happening because hunger feels like something is happening.

No coffee enemas. No established elimination benefit, real documented risks.

No “detox teas.” Most are stimulant laxatives in a floral bag. Bowel regularity built on senna is not drainage; it is dependence with better branding.

No 30-day protocol. The habits in this guide are the maintenance state of a body that eliminates well. A calendar with an end date is the wrong shape for this job.

What a day looks like on this pattern

- **Breakfast:** eggs with sauteed greens and onions; oatmeal with chia and berries on the side. Protein, alliums, fiber, color.
- **Lunch:** a big salad with a palm of salmon or lentils, a handful of broccoli sprouts, olive oil, and whatever vegetables are around. Crucifer, protein, fiber.
- **Dinner:** any protein, two vegetables where one is cruciferous, beans or a whole grain. Water through the day to pale-straw urine.
- **The weekly rhythm:** fish twice from the low-mercury list, legumes most days, sprouts growing on the windowsill in rotation, food reheated in glass.

Nothing exotic, which is the point. The exotic version was never the supported version.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your actual exposure audit, your labs, and your history.

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Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

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Read the companion documents alongside this one. **Understanding Your Toxin and Detox Labs** covers which tests are real and the one to refuse, and **The Detox Supplement Guide** covers what genuinely helps beyond the plate.

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All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Detox-Kit-Citation-Ledger.md`. The FDA/EPA fish consumption advice is standing public health guidance cited by name above.

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Educational content only. Not medical advice. Not a diagnosis. Not a reason to stop or refuse any medication. Suspected acute poisoning is an emergency: Poison Control 1-800-222-1222.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Chelation is never a do-it-yourself project. Not oral, not “gentle,” not “natural.” The section below explains what has actually happened to people, including children, when it was treated as one.

Binders interfere with medications. Anything designed to bind compounds in your gut will also bind drugs and nutrients taken near it. A binder does not know the difference between a toxin and your thyroid medication. If one is genuinely indicated for you, it goes 1 to 2 hours away from food, supplements, and medication, every time. If you take any prescription medication, that interaction is real and it is yours to manage with your prescriber, not to discover by accident.

Suspected acute poisoning is an emergency. Poison Control: 1-800-222-1222. No supplement has a role in that situation.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

Start with the uncomfortable fact about this product category

The detox supplement aisle has a liver problem, and it is not the one it advertises.

Herbal and dietary supplements now account for roughly 20 percent of drug-induced liver injury cases in the United States, per a research symposium review sponsored by the American Association for the Study of Liver Diseases and the NIH. The major implicated agents: anabolic steroids sold as bodybuilding products, **green tea extract**, and multi-ingredient nutritional supplements, the exact format most “detox blends” ship in, where the responsible ingredient usually cannot even be identified afterward (PMID 27677775).

Sit with that for a moment, because it reframes the whole category. The organ detox products claim to support is the organ this product category most often injures. Concentrated green tea extract, a fixture of cleanse formulas, is one of the most consistently implicated single ingredients. Whole green tea as a beverage is not the concern; the concentrated extract in pills is.

The rule this buys you: skip multi-ingredient “detox,” “cleanse,” and “liver support” blends entirely. Every ingredient you take should be one you chose on purpose, at a dose you know, for a reason you can say out loud. That rule alone outperforms most of what this aisle sells.

Tier 1: what actually has human evidence

Glutathione: the honest version of the strongest story

Glutathione is the master Phase II conjugate, and intracellular depletion under oxidative load is real physiology, which is why it headlines this tier.

What a randomized trial actually found: six months of daily oral glutathione at 250 or 1,000 mg in 54 healthy adults raised body stores measurably, 30 to 35 percent in blood cells, plasma, and lymphocytes at the higher dose, with a reduction in oxidative stress markers and a dose-and-time-dependent pattern. Levels returned to baseline within a month of stopping (PMID 24791752).

Who paid for it, because this kit checks that everywhere: the trial was funded by the manufacturer of the branded glutathione ingredient it tested. That does not make the measurements wrong, and the finding is mechanistically plausible, but it is the only trial of its size on this question and the sponsor profits from the answer. Weight it accordingly.

What that does and does not establish. It establishes that oral glutathione raises glutathione status, which for years was genuinely in doubt. It does not establish that raising glutathione in a healthy person improves symptoms or clears toxicants faster; those outcome trials have not been done. So: a reasonable, evidence-grounded choice when a measured RBC glutathione is low (the lab guide covers that marker), an act of faith when it is not measured.

NAC: the precursor with a pharmacy pedigree

N-acetylcysteine supplies cysteine, the rate-limiting ingredient for the body's own glutathione synthesis. Its safety record is unusually deep because it is also a real medicine, used for decades in acetaminophen overdose. An evidence review states the honest overall picture plainly: well-established safety and mechanism, with clinical trial effectiveness across the various conditions studied still limited (PMID 34208683).

Practical notes: typical studied doses run 600 to 1,800 mg daily. The classic complaint is the sulfur smell and mild stomach upset. If you take nitroglycerin for angina, NAC can intensify its blood-pressure-lowering effect; that combination belongs with your prescriber.

Sulforaphane: prefer the food, and here is the honest supplement note

The randomized human trials showing increased excretion of real airborne toxicant conjugates used broccoli sprout preparations, not pills (PMID 24913818, 31268126). The dose-ranging trial's practical gift is that even one-fifth of the original dose still worked, which puts a daily handful of home-grown sprouts within range of the studied territory.

Supplements are the fallback, and the category has a known chemistry problem: sulforaphane yield varies widely between products depending on whether the converting enzyme (myrosinase) survives processing. If you go the capsule route, choose a product that states its sulforaphane or glucoraphanin yield and carries third-party testing. Or grow sprouts for pennies and skip the problem.

Tier 2: popular, with weaker evidence, stated honestly

Milk thistle (silymarin)

The most famous liver herb has a genuinely thin outcome record. The Cochrane review of milk thistle for alcoholic and viral liver disease found no firm evidence of benefit on mortality or liver histology, with the positive signals concentrated in the lowest-quality trials (PMID 17943794). A later meta-analysis found statistically significant reductions in liver enzymes that the authors themselves judged too small to be clinically relevant (PMID 28785154).

The honest position: reasonably safe at studied doses, unlikely to hurt, repeatedly failing to show it meaningfully helps. If a normal-livered reader wants a liver herb because it feels supportive, this is the least concerning choice in the aisle. That is the strongest sentence the evidence permits.

Chlorella, cilantro, and the “gentle binder” shelf

Widely sold for metal detox on the strength of animal and in vitro work; controlled human evidence that they reduce body burden is essentially absent. This guide will not build a protocol on that. If future trials change the picture, this section changes with it.

Activated charcoal

A real emergency-room drug for certain acute ingestions, repurposed by marketing into a daily habit. Taken routinely it binds indiscriminately, medications and nutrients included, and no controlled human evidence supports routine use for chronic “toxin” reduction. Reserve charcoal for the emergency department, where it actually belongs.

What not to do, and what has actually happened

Chelation without a confirmed diagnosis

Chelating drugs are real pharmaceuticals that mobilize stored metal into active circulation. Used on the wrong person, or confused between agents, they have killed: documented deaths from sudden hypocalcemia after the wrong EDTA formulation, including children (PMID 16882789). The Cochrane review of chelation for autism, the use behind some of those deaths, found no clinical trial evidence of benefit and concluded the risks outweigh any proven benefit (PMID 26106752).

If legitimate unprovoked testing confirms clinically significant toxicity, chelation is real medicine with a real specialist and real monitoring. Everything short of that, including every oral “metal detox” product sold without a diagnosis, is risk without an established benefit.

High-dose selenium

Selenium's window is narrow, and the antioxidant enzymes it serves are already saturated at adequate status. Supplement only against a measured level (the lab guide's functional target is 85 to 140 ug/L), not on principle.

High-dose vitamin C, for one specific group

If you have ever formed a calcium oxalate kidney stone, know that ascorbate is a documented precursor of endogenous oxalate production (PMID 30566003). Normal dietary and modest supplemental amounts are one thing; gram-level daily dosing is a conversation for your physician with your stone history on the table.

The supporting cast, briefly

A basic B-complex covers the Phase I and methylation cofactors when the diet is inconsistent. Unglamorous, cheap, reasonable.

Magnesium, most evenings, is the practical lever when bowel regularity is the missing foundation; magnesium citrate or glycinate at modest doses draws water into the bowel and reliably helps. Bowel regularity is a precondition for everything else in this kit, so this humble entry outranks most of the aisle.

Zinc, only against a measured plasma level, drawn when you are well, since acute illness redistributes zinc and fakes deficiency.

Sauna and sweating are not supplements but get asked about here: a small human monitoring study found certain elements detectable in sweat, sometimes when blood was undetectable (PMID 21057782). Preliminary, small, and honest words for it; enjoyable and plausibly additive, proven for nothing, and not a route PFAS takes at all.

What I deliberately left out

Every multi-ingredient cleanse and flush, coffee enemas, “detox foot pads,” zeolite sprays, and every product whose mechanism is described in energy language rather than chemistry. The bar for a document sold to a stranger I have never examined is higher than the bar for a conversation in my office, and none of those clear it. The strongest things in this entire kit remain free: the exposure audit, the fish swap, the sprouts on your windowsill, fiber, water, sleep, and movement.

Where to find quality versions of these

Quality varies more in this category than almost anywhere in the supplement world, and third-party testing (NSF, USP, or equivalent independent certification) matters more than brand marketing. That guidance stands whether you buy from us or anywhere else.

I keep a screened dispensary through Fullscript, which ships directly to you. I hold no physical inventory. It carries practitioner-grade lines with the testing practices described above:

us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Nothing here was included, excluded, ranked, or dosed based on what it earns, and the highest-value items in this guide, the exposure audit, broccoli sprouts, fiber, and water, earn us nothing.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your exposure history, your labs, and your medication list, which is exactly what the capacity markers in the lab guide are for.

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Read the companion documents alongside this one. **Understanding Your Toxin and Detox Labs** covers the capacity markers that turn this guide from guessing into responding, and **The Detox Diet Guide** covers the food-first levers that outperform most of this page.

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All references were verified against live PubMed records on 2026-08-09, including a publication type and corrections check for retractions and errata. Verification detail is documented in [Detox-Kit-Citation-Ledger.md](#).

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